

Afsana Bhuiyan

Software Engineer | Python, Typescript, C#/.NET

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Work Experience

Margin Research

New York, NY

Software Engineer (Contract)

February '26 – June '26

- Deployed Reagent, a SaaS code-analysis platform, from prototype to v1 for government customers within three months, building the Python backend and React frontend from the ground up.
- Designed and built a five-stage backend processing pipeline using Celery and Redis/Valkey, coordinating asynchronous analysis and indexing results in Elasticsearch for search and retrieval.
- Built a reusable TypeScript visualization tool for graph-based analysis, supporting multiple data formats and per-node analytics; adopted across two internal projects.
- Containerized application services with Docker for CI/CD deployment and wrote automated tests covering task processing and failure handling.

Pieces

New York, NY

Software Engineer Intern

September '25 – December '25

- Built a full-stack application using Flask, JavaScript, and the Gemini API to generate synthetic patient records for training and testing the company's AI document-summarization product.
- Designed structured data models for longitudinal patient records, collaborating with physician consultants to validate data quality and clinical realism.
- Engineered prompts and a resilient data-processing pipeline that detected and repaired malformed LLM responses before validation, enabling reliable generation of 200+ clinical notes per patient.
- Implemented configurable batch processing and structured JSON exports for downstream model training, with an interface for reviewing generated patient records.

COOP Careers

New York, NY

Data Analytics Fellow

January '25 – June '25

- Built Tableau dashboards and analyzed a 508-movie dataset using Python, pandas, and NumPy to identify performance trends across markets, franchises, and genres.

Projects

[El Principito](#) - Language Learning Companion | C#, ASP.NET Core, Entity Framework Core, SQLite

August '26 – Present

- Building a language-learning reading application for Spanish learners, with an initial dataset of five Spanish children's books stored as JSON; established the RESTful API and implemented the core book data model.
- Currently implementing API endpoints and relational database structure to support book retrieval, vocabulary lookup, and persistent language-learning data.

[CareTrack](#) - Patient Waitingroom Dashboard | React.js, Node.js, PostgreSQL, Azure

March '26 – Present

- Built a full-stack clinical workflow platform using React, Express, and PostgreSQL, automating physician documentation deadlines with scheduled task generation, role-based access control, audit logging, and GitHub Actions CI/CD on Azure.

Education

Rensselaer Polytechnic Institute | Troy, NY

June '24

B.S. Major in Computer Science, Data and AI Specialization

Skills

- **Languages:** Python, TypeScript, JavaScript, C#, SQL, Java, C++
- **Frameworks & Tools:** React.js, FastAPI, Flask, Node.js, ASP.NET Core, Pytest, JUnit, PostgreSQL, SQLite, Elasticsearch, Docker, Celery, Redis/Valkey, Git, Linux